



AUTOMATE OR DIE 2026 / DAY 2

SUPCOM / Oussama Akir / solo participant

See the consequences before changing the workshop.

CableTwin converts one line stoppage into three verified recovery routes, adds local learned guidance, and keeps the decision with the human manager.

LIVE ISOMETRIC TWIN

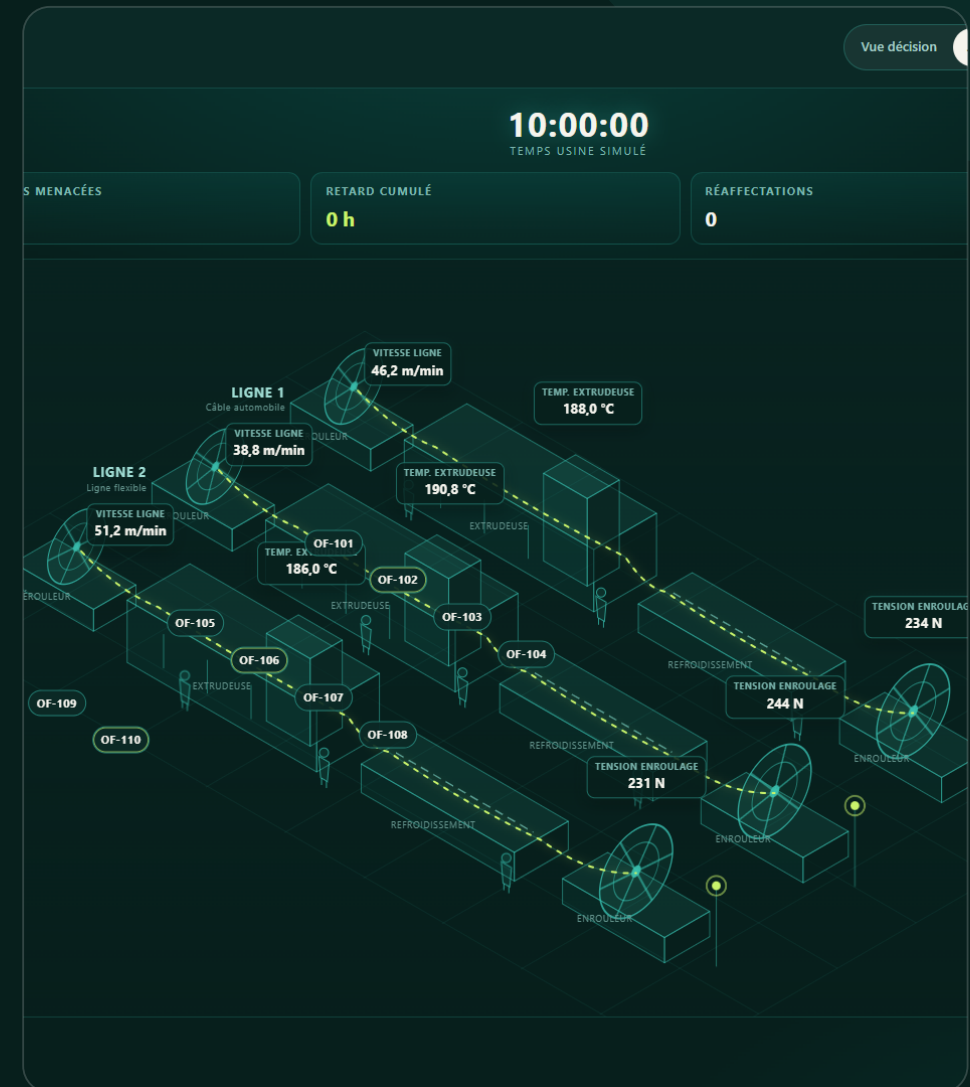
LOCAL ML

HUMAN APPROVAL

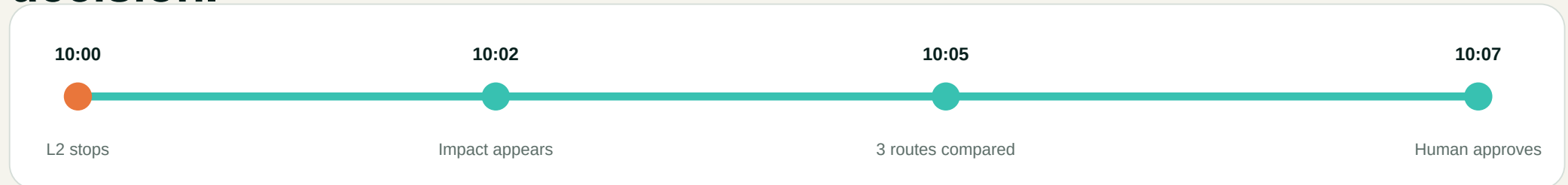
UPDATED CONCEPT NOTE

Includes complete 9-block Business Model Canvas

Live product / hackaton-automate-or-die.vercel.app/#twin



At 10:00, Line 2 stops. The schedule becomes a business decision.



620 min

delay if all orders stay on their original line

7 / 10

orders on time under Stability

180 min

projected overtime under Stability

188 km

completed by 18:00 under Stability

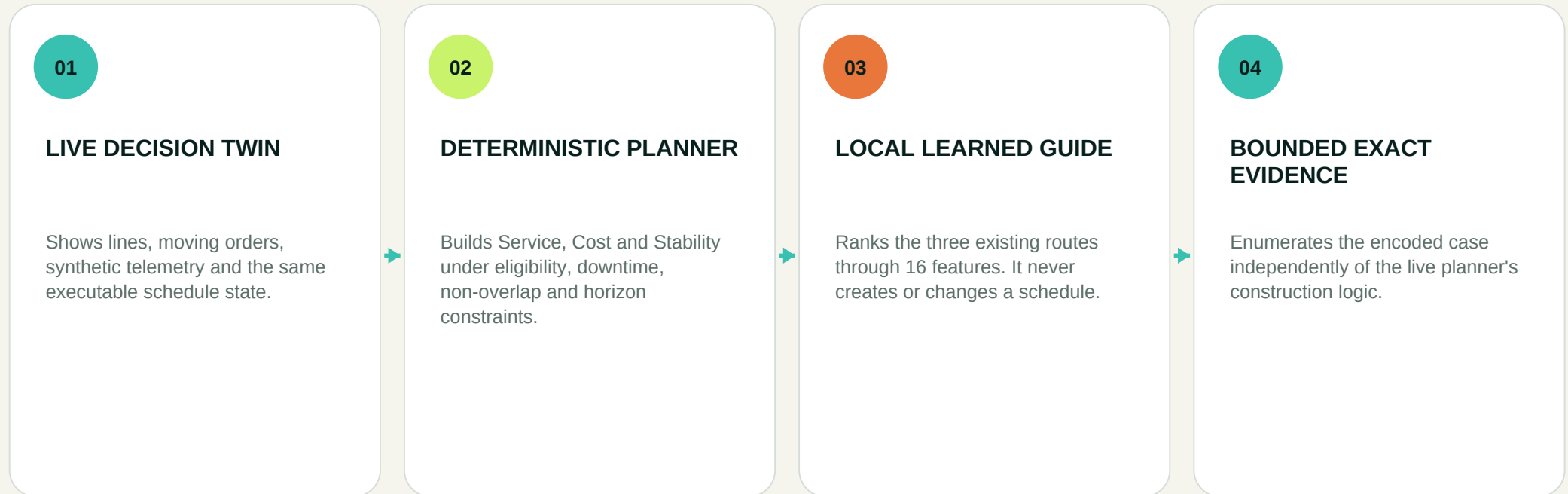
WHAT CABLETWIN CHANGES

- **Three feasible routes** replace one opaque recommendation.
- **One shared view** aligns production, service and management.
- **Explicit approval** preserves accountability.

WHAT CABLETWIN DOES NOT DO

- No machine, PLC, ERP, MES or APS control.
- No external AI call and no real factory data in the demo.
- No promised ROI, quotation or autonomous decision.

Intelligence is separated so the jury can audit every claim.



THE FINAL GATE IS HUMAN



The model may suggest **Cost**. The production manager may still choose **Service**, preview the revised schedule, approve it and create an auditable event.

Current deployment boundary: local / offline-capable / read-only / no write-back

Three routes. Different priorities. One transparent trade-off.

ROUTE	ON TIME	DELAY	OVERTIME	MOVES	18:00 OUTPUT	SYNTHETIC COST
Service	8 / 10	140 min	30 min	3	216 km	1,971.15 DT
Cost	8 / 10	170 min	60 min	2	216 km	1,931.45 DT
Stability	7 / 10	620 min	180 min	0	188 km	3,862.20 DT

9 / 9

planner and workflow checks

Reported separately from the learned layer.

5 / 5

local recommender checks

687 twin-generated synthetic incidents / 16 features.

93.6%

held-out synthetic accuracy

95.7% train / Cost about 79% on the canonical incident.

17,856

bounded candidates enumerated

10,440 feasible / 3 unique policy optima.

EVIDENCE BOUNDARY

Every input, order, operator, customer, sensor value, cost coefficient and result is synthetic. DT values are comparison outputs - not quotations, factory savings or commercial ROI. The exact verifier proves only the encoded demonstration.

Business Model Canvas

All pricing, revenue and partner assumptions remain to be validated.

KEY PARTNERS

- Cable and electrical-equipment design partners
- ERP / MES / APS integration specialists
- Industrial associations and applied research
- Cybersecurity and on-premise infrastructure specialists
- Targets only - no confirmed partnership claimed

KEY ACTIVITIES

- Map planner-validated constraints
- Configure read-only ingestion
- Validate feasibility, security and usability
- Measure shadow-pilot outcomes
- Maintain audit, rollback and governance

VALUE PROPOSITIONS

- Three constraint-feasible routes before any workshop change
- Visible delay, delivery, overtime, movement and output trade-offs
- Deterministic planning plus explainable learned guidance
- Human preview, approval and traceability
- Local, offline-capable and read-only first deployment
- Complement existing planning systems instead of replacing them

CUSTOMER RELATIONSHIPS

- Constraint co-design with planners
- Guided setup and onboarding
- Joint pilot evidence review
- Local support and monitoring
- Explicit deterministic fallback

CUSTOMER SEGMENTS

- Primary user: production manager or planner
- Buyer hypothesis: plant or operations director
- Secondary: service, engineering, finance and IT
- Beachhead hypothesis: Tunisian cable and electrical-equipment plants
- Multi-line workshops with shared product eligibility
- No signed customer or pilot claimed

KEY RESOURCES

- Constraint and KPI model
- Synthetic decision twin
- Local recommender pipeline
- Bounded exact-verification harness
- Plant co-design and solo-builder capacity

CHANNELS

- Direct operational discovery
- Workshop demonstrations
- Read-only shadow pilots
- Industrial challenge networks
- Complementary integration partners

COST STRUCTURE - QUALITATIVE

- Product engineering, integration and support
- Local deployment, security hardening and validation
- Pilot discovery, training and evidence collection
- Model monitoring and periodic plant-specific evaluation
- No financial estimate claimed

REVENUE STREAMS - HYPOTHESES ONLY

- Paid read-only pilot after scope agreement
- Configuration and integration services
- Per-site subscription only after measured pilot value
- Pricing, revenue and conversion assumptions remain unvalidated

Start read-only. Earn trust with evidence. Learn locally.

WEEK 1

Map

Validate line eligibility, durations, calendars, incident records and KPI definitions with planners.

WEEKS 2-3

Replay

Run representative historical incidents beside actual decisions. No write-back and no machine control.

WEEKS 4-5

Evaluate

Compare assessment time, delay, output, on-time rate, overtime, moves, violations and usability.

WEEK 6

Decide

Review evidence, deployment fit and safeguards before any commercial or integration commitment.

LEARNED LAYER GATE

Retrain only after the site has sufficient representative real incident history.

- Training and inference remain on-site; plant data does not leave.
- Use a separate representative holdout for plant-specific evaluation.
- If the learned layer underperforms, keep the deterministic three-route planner.

CURRENT MATURITY

PHASE 1	Decision concept
PHASE 2	Accepted end-to-end proof
DAY 2	Live twin + local guidance

Next ask: one industrial design partner for a bounded read-only shadow pilot.